

One-meter parking at every car length

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Status and scope

This is an ordinary-mathematics proof with a companion Lean formalization in root Main.lean. The formalization proves the actual-count generating function, scalar recurrence, Lucas range, first correction, and boundary cases by division-free finite-state elimination, not a line-by-line formalization of this note's q /geometric-sum presentation. A bounded literature audit does not establish priority. The note gives an explicit all-length consequence of the last-outcome decomposition used in the source paper; it makes no claim that the parking model or starting state decomposition is new.

Fix an integer $n \geq 1$. There are n spaces and every car has meter duration $t = 1$. Write $[n] = \{1, \dots, n\}$. A car has an ordered preference $a \in [n]$ and parks at the first available space at or to the right of a , failing if there is none. The previous car departs *after* the next car parks, so a vacated space may be reused by a later car. Let $F_m(n)$ be the number of successful ordered preference lists of length m , with $F_0(n) = 1$, and put

$$G_n(x) = \sum_{m \geq 0} F_m(n) x^m.$$

The all-length formula

Theorem. For every $n \geq 1$,

$$G_n(x) = \frac{(1+x)^n}{(1-nx+x^2)(1+x)^n - x^{n+2}}.$$

For $2 \leq j \leq n$, define

$$c_{n,j} = n \binom{n}{j-1} - \binom{n}{j} - \binom{n}{j-2}.$$

Then

$$F_m(n) = \sum_{j=2}^n c_{n,j} F_{m-j}(n) \quad (m \geq n+1).$$

Together with $F_0(n) = 1$, $F_1(n) = n$, and

$$F_m(n) = nF_{m-1}(n) - F_{m-2}(n) \quad (2 \leq m \leq n),$$

this gives sufficient initial data. When $n = 1$, the displayed sum is empty. No assertion about minimal recurrence order is intended.

Proof

For a nonempty successful prefix of length r , precisely its last car remains immediately before the next search. Let $z_i(r)$ count the prefixes whose last outcome is space i . If the next preference is a , then, from previous outcome i :

- if $a \neq i$, the next outcome is a ;
- if $a = i < n$, the next outcome is $i + 1$;
- if $a = i = n$, parking fails.

Thus the transition from outcome i to outcome j has multiplicity

$$w_{ij} = \mathbf{1}_{j \neq i} + \mathbf{1}_{j=i+1}.$$

The multiplicity is importantly two when $j = i + 1$: the distinct preferences i and $i + 1$ both produce that outcome. Each extension has a unique prefix and appended preference, so these transitions count ordered lists exactly once. Consequently, for every $r \geq 1$,

$$z_j(r + 1) = F_r(n) - z_j(r) + z_{j-1}(r), \quad z_0(r) = 0, \quad z_j(1) = 1,$$

and $F_r(n) = \sum_{j=1}^n z_j(r)$. This is the source paper's last-outcome decomposition. Its local justification applies at every length, rather than only in the range used there to obtain the bounded Lucas recurrence.

Work in the formal-series ring $\mathbb{Z}[[x]]$. Set

$$Z_j = \sum_{r \geq 1} z_j(r) x^r, \quad Z_0 = 0, \quad q = \frac{x}{1+x}.$$

Multiplying the state equation by x^{r+1} and summing over $r \geq 1$ gives

$$Z_j - x = x(G_n - 1) - xZ_j + xZ_{j-1}, \quad Z_j = q(G_n + Z_{j-1}) = G_n \sum_{\ell=1}^j q^\ell.$$

After summing over j , this becomes $G_n - 1 = G_n S_n$, where

$$S_n = \sum_{\ell=1}^n (n + 1 - \ell) q^\ell.$$

The finite identity

$$(1 - q)^2 S_n = nq - (n + 1)q^2 + q^{n+2}$$

and $1 - q = (1 + x)^{-1}$ give

$$S_n = nx - x^2 + \frac{x^{n+2}}{(1+x)^n}.$$

Solving $G_n - 1 = G_n S_n$ proves the generating function. Every series inverted here has constant term 1, so no analytic convergence is being used.

Let

$$D_n = (1 - nx + x^2)(1 + x)^n - x^{n+2}.$$

The coefficients of x , x^{n+1} , and x^{n+2} in D_n cancel, and hence

$$D_n = 1 - \sum_{j=2}^n c_{n,j} x^j.$$

Since $D_n G_n = (1+x)^n$, coefficient extraction above degree n gives the stated recurrence. It begins at $m = n+1$, not $m = n$, because the numerator has coefficient 1 in degree n .

Finally,

$$(1 - nx + x^2)G_n = 1 + x^{n+2}(1+x)^{-n}G_n.$$

The series multiplying x^{n+2} has constant term 1. Thus the Lucas recurrence holds through $m = n+1$, and its first correction is exactly

$$\boxed{F_{n+2}(n) = nF_{n+1}(n) - F_n(n) + 1.} \quad \square$$

Boundary cases and examples

The generating functions for the first three values of n are

$$G_1 = 1 + x, \quad G_2 = \frac{(1+x)^2}{1-2x^2}, \quad G_3 = \frac{(1+x)^3}{1-5x^2-5x^3}.$$

For $n = 3$, the counts beginning at length zero are

$$1, 3, 8, 21, 55, 145.$$

In particular, $145 = 3 \cdot 55 - 21 + 1$, not 144. This shows why the stated Lucas recurrence cannot be continued without its length bound. The cases $n = 1$ and $n = 2$ are already covered (the first is immediate and the second is covered at all lengths in the source). Thus the genuinely additional range of the all-length formula is $n \geq 3$ and $m \geq n+2$.

Relation to the source and literature boundary

Daugherty, Harris, Klein, and McClinton, *Metered Parking Functions*, *Integers* **25** (2025), A73, pose Problem 2 on p. 32: find a recursive or closed formula when $m > n$. Their Theorem 2 (pp. 14–15) proves the Lucas recurrence only for $2 \leq m \leq n+1$, and Proposition 6 (pp. 17–18) already treats all lengths when $n = 2$. The explicit all-length elimination above answers the stated Problem 2 mathematically by reusing and extending the local applicability of the paper’s last-outcome decomposition; it does not claim a new model or state equation.

No exact all-length formula of this form was identified in the sources inspected during the bounded check, but that is not a first-solution or priority claim. The paper is also available as arXiv:2406.12941. OEIS A372817 is a useful data table; its entry includes 145, which is incompatible with continuing the bounded Lucas recurrence without qualification. No statement here assigns a cause to that presentation.

Finite validation and disclosure

A finite, definition-level check examined 586023 ordered inputs for $n = 1, \dots, 5$, $m = 0, \dots, 8$, and $t = 1$, using two original simulators; the resulting 45 count rows agreed with the formula. A separate state/scalar comparison covered $n = 1, \dots, 20$ and $m = 0, \dots, 40$. This finite evidence is not a premise of the proof and is not evidence of novelty.

Sources

1. Spencer Daugherty, Pamela E. Harris, Ian Klein, and Matt McClinton, “Metered Parking Functions,” *Integers* **25** (2025), A73, Problem 2 p. 32, Theorem 2 p. 15 (supporting argument pp. 14–15), and Proposition 6 p. 18 (introduction p. 17). DOI: 10.5281/zenodo.16881806. <https://math.colgate.edu/~integers/z73/z73.pdf>. arXiv:2406.12941.
2. OEIS Foundation Inc., “A372817,” <https://oeis.org/A372817>.